

Baptiste Lerat

Quantum Physicist

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github.com/balerat

Experience

Atos / Eviden

Paris, France

Research Intern – Quantum Computing

Dec 2023 – Jul 2024

- Using Tensor Network to simulate analog quantum computers with quantum trajectories techniques.
- Developed simulation tools in python and C++ to use quantum computers on condensed matter problems.
- Researched quantum advantages in simulating quantum phase transition and critical exponent on Mott insulator - superfluid transition for Rydberg atom quantum processors.

CNRS

Lyon, France

Research Intern – Nanotechnologies and Photonics

May 2022 – Aug 2022

- Investigated the Klein paradox in an optical fishbone structure, relativistic quantum tunneling effects on photons.
- Performed simulations using RSoft and MATLAB to characterize material properties, focusing on parameters affecting photon tunneling and resonance on top of conducting a literature review on the topic.

Projects

Imperial College of London

London, UK

Master Thesis – Wigner Tomography of Transmon qubits

May 2023 – Oct 2023

- Python simulation of two transmon coupling systems via a microwave resonator for parameter optimization with Bayesian optimization. Analysis of the shifts induced in the system with Mathematica.
- Designed a Wigner Tomography experiment protocol of transmon qubit for possible patented application.
- Researched the mathematical formalism of transmon qubit coupling with Laplacian discretization methods.

Centrale Lyon

Lyon, France

Research project – Strong coupling of light and matter

2021 – 2022

- Studied the strong coupling of light in order to create cavity polaritons using perovskite materials.
- Designed dielectric Bragg mirror and cavity spectrum on MATLAB. Fabrication of the cavity in cleanroom environments with spin coating technique. and creation of an optical table to observe and quantify the coupling of the polaritons.

Education

UNIVERSITY OF CAMBRIDGE

Cambridge, UK

PhD in Many-body Quantum physics

Oct 2024 – Present

Research focused on many-body physics at AMOP group. Many-body Localization in quasicrystal structure with cold atoms.

IMPERIAL COLLEGE

London, UK

MSc in Physics with Quantum dynamics – 3.8/4 GPA, Distinction

2022 – 2023

Coursework: Quantum Information, Quantum Optics, Nonlinear Optics, Atomic structure, Cold Atomic Systems, Hybrid Quantum Systems, Quantum theory of matter, Quantum dynamics, Quantum error correction.

CENTRALE LYON – Bsc and MSc in Engineering – 3.86/4 GPA, Distinction

Lyon, France 2020 – 2024

Lycée Chaptal – CPGE « classes préparatoires » MP*

Paris, France 2018 - 2020

Skills and Interests

Programming: Python (NumPy, SciPy, Pandas, Qutip, Qiskit, TensorFlow), C++, MATLAB, Mathematica

Machine learning certification - Stanford & Deeplearning.ai

Coursera formation: Neural Network, TensorFlow, Clustering, XGBoost, K-means, Decision trees, Linear regression

Math and Physics Club – Founder - Centrale Lyon: Founded the first math and physics club in Centrale Lyon with the objective of giving students interest in fundamental sciences and testing their problem-solving skills.

Physics supervisor - University of Cambridge : Supervising undergraduate students in advanced quantum mechanics, providing guidance on problem sets. Offering one-on-one and group sessions to support students' understanding of complex topics.